

Course: B.Tech in Computer Engineering and CSE

Semester: I

Subject Code & Name: Digital Electronics (23UD1245ES106)

Max Marks: 60

Date: 24th January 2026

Duration: 3 Hr.

Instructions to the Students:

- 1 Each question carries 12 marks.
- 2 Question No. 1 will be compulsory and include objective-type questions.
- 3 Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- 4 The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- 5 Use of non-programmable scientific calculators is allowed.
- 6 Assume suitable data wherever necessary and mention it clearly.

Marks

- Q. 1 Objective type questions. (Compulsory Question) 12**
- 1 Which of the following represents a negative edge-triggered clock pulse? 1
A) High level signal B) Transition from 0 to 1
☒ C) Transition from 1 to 0 D) Low level signal
 - 2 Which of the following is NOT a standard type of shift register? 1
A) SISO (Serial-In Serial-Out) B) SIPO (Serial-In Parallel-Out)
☒ C) FIFO (First-In First-Out) D) PIPO (Parallel-In Parallel-Out)
 - 3 The 2's complement of the binary number (1010) is 1
A) 0101 ☒ B) 0110 C) 1100 D) 1011
 - 4 Which gate is known as the "Universal Gate" because it can implement any other Boolean function? 1
A) AND B) OR ☒ C) NAND D) XOR
 - 5 A 4-variable K-map contains how many total cells? 1
A) 4 B) 8 ☒ C) 16 D) 32
 - 6 Which code is specifically designed to detect and correct single-bit errors in data? 1
A) Gray Code ☒ B) Hamming Code C) BCD Code D) Excess-3 Code
 - 7 How many inputs does a Half Adder have? 1
A) 1 ☒ B) 2 C) 3 D) 4
 - 8 Combinational circuit selects one of many inputs and steer it to a single output _____. 1
A) Decoder ☒ B) Demultiplexer C) Multiplexer D) Priority Encoder
 - 9 In an asynchronous (ripple) counter, the external clock is applied to: 1
☒ A) All flip-flops simultaneously B) Only the first flip-flop
C) Only the last flip-flop D) No flip-flops

- 10 What is the maximum decimal value that can be represented by a 4-bit binary number? 1
 A) 7 B) 8 ☒ C) 15 D) 16
- 11 Which flip-flop is specifically designed to eliminate the "race around" condition? 1
 A) SR Flip-flop B) D Flip-flop ☒ C) Master-Slave JK Flip-flop D) T Flip-flop
- 12 A 1-to-4 Demultiplexer (Demux) has how many select lines? 1
 A) 1 ☒ B) 2 C) 3 D) 4

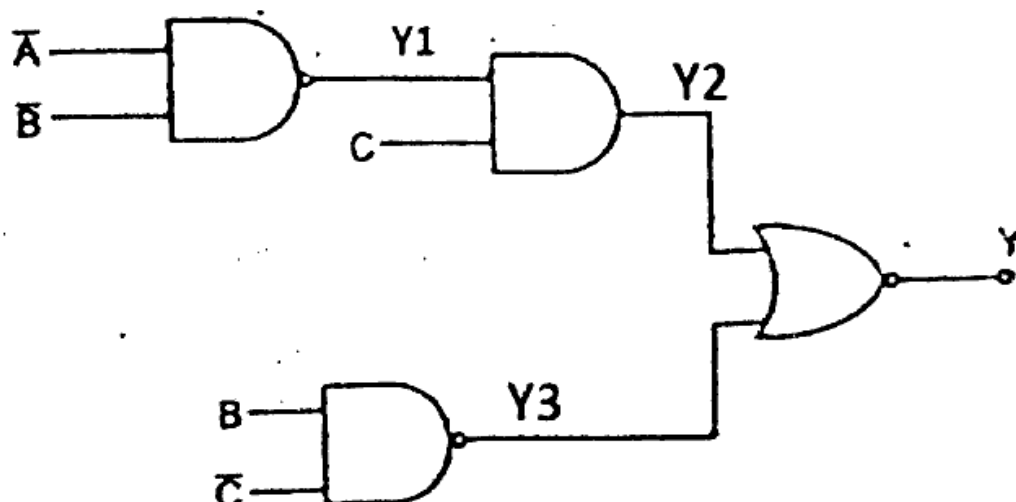
Q. 2 Solve the following.

12

A) For the logic circuit shown below:

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- ☒ a) Derive the intermediate Boolean expressions for the values of Y1, Y2, and Y3
- ☒ b) Derive the final Boolean expression for the output Y
- ☒ c) Simplify the final expression using Boolean laws and draw the circuit diagram of the reduced expression. <https://www.pyqonline.com>



- ☒ B) State and prove both of De Morgan's Theorems (using Truth Tables). Simplify the following logic expression using Boolean laws $Y = AB + A\bar{B} + \bar{A}C + \bar{A}\bar{C}$. 6

Q.3 Solve the following.

12

A) Perform the following numerical operations:

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- ☒ a) Perform the subtraction: $(50)_{16} - (2A)_{16}$ using the 2's complement method.
- ☒ b) Perform the BCD addition of $(25)_{10} + (36)_{10}$. Show all steps and the correction process.

B) ☒ a) Convert the binary number $(11010011)_2$ into its equivalent Octal, Hexadecimal, and Gray Code representations.

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- ☒ b) A 7-bit Hamming code is received as 1110101. Using even parity, determine the bit position where the error is present.

Q. 4 Solve Any Two of the following.

12

- ☒ A) a) Find the reduced form of the following function using K-map:

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$$F(W, X, Y, Z) = \sum m(0, 7, 8, 9, 10, 12) + d(2, 5, 13).$$

- b) Minimize the following expression using K-map:

$$f(A, B, C, D) = \prod M(0, 1, 4, 5, 7, 8, 9, 12, 13, 15).$$

- ☒ B) Design a 3-bit Majority Detector circuit. The system has three data inputs (A, B, C) and a single output (Y). The output Y is logic high ('1') only when at least two of the input bits are logic high. For this system:

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- Construct the Truth Table.
- Obtain the simplified Sum of Products (SOP) expression using a K-map.
- Draw the Logic Implementation using basic gates.

- C) Simplify the following logic equation using K-map: $ABC + \bar{B}CD + \bar{A}BC$. Realize the final simplified expression using NAND gates only.

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Q.5 Solve Any Two of the following.

12

- ☒ A) Design a Half Adder circuit. Provide the following:

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- Block Diagram and Truth Table.
- K-maps for the derivation of Boolean expressions for Sum and Carry.
- Final logic circuit diagram using basic gates.

- ☒ B) Construct the Truth Table for a BCD to 7-Segment Decoder. Using a K-map, obtain the simplified logical expression for segments 'a' and 'b' only.

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- C) Define a Demultiplexer. Explain the working and truth table of a 1:4 Demultiplexer and draw its logic implementation using basic gates.

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Q. 6 Solve Any Two of the following.

12

- A) What is an Asynchronous (Ripple) Counter? Explain a 4-bit asynchronous UP (ripple) counter with neat diagram and working.

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- B) Define register and shift register. Explain SISO, SIPO, PISO, PIPO shift registers with block diagram and applications.

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- C) Explain Race around condition. Why does it occur. How is it avoided and explain Master slave flip flop.

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*** End ***